L1 : MORE ABOUT FACTORIZATION (A)

Overview:

1 3 5 8

2 4 6 7

Which one(s) are the factors of 8?

8 = 1 x 8 = (1)(8)
8 = 2 x 4 = (2)(4)

Factorize = Find the factors

∴答案格式一定係 (   )(   )

Checkpoint 1

$a^2 + 2ab + b^2 = (a+b)^2$

Step 1:

$4a^2 + 16a + 16$

$= (2)^2(a)^2 + 2(8a) + (4)^2$

Step 2:

$= (2a)^2 + 2(2a)(4) + (4)^2$

$= [(2a) + 4]^2$

$= (2a+4)^2$

Checkpoint 2

$a^2 - b^2 = (a+b)(a-b)$

Step 1:

$16a^2 - 9b^2$

$= 4^2a^2 - 3^2b^2$

$= (4a)^2 - (3b)^2$

Step 2:

$= [(\quad) + (\quad)]\,[(\quad) - (\quad)]$

$= [(4a) + (3b)][(4a) - (3b)]$

$= (4a+3b)(4a-3b)$

Unit Education

Let's get started!

Factorize the following expressions.

1. $a^2 + 4a + 4$

2. $4c^2 - 12c + 9$

3. $25d^2 + 20d + 4$

4. $9b^2 - 6b + 1$

5. $16m^2 + 8mn + n^2$

6. $p^2 - 10pq + 25q^2$

7. $16a^2 - 1$

8. $36m^2 - 25n^2$

9. $4a^4 - b^2c^2$

10. $32x^2 - 50y^2$

11. (a) Factorize $16x^2 - 8x + 1$.
    (b) Hence, factorize $16y^4 - 8y^2 + 1$.

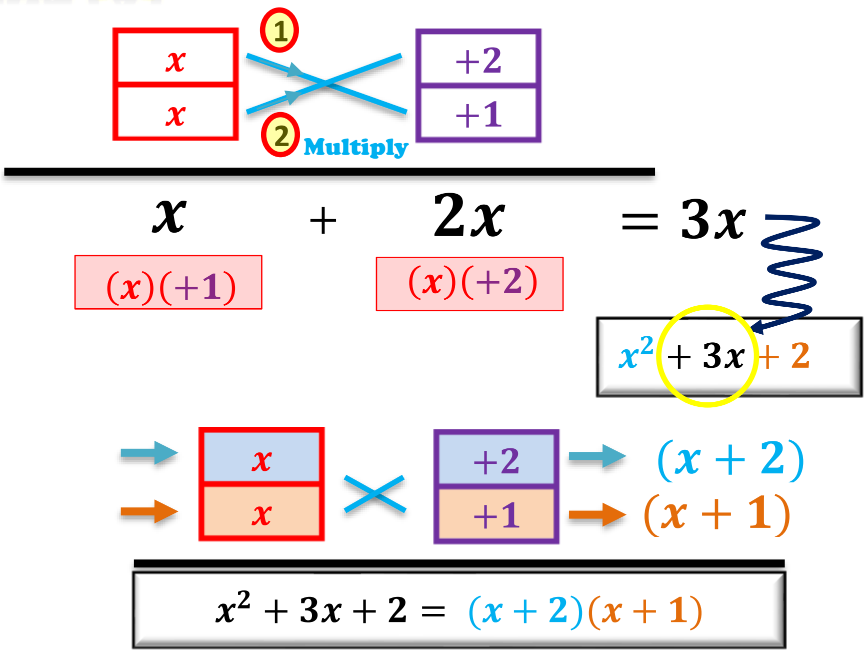
12. (a) Factorize $5x^2 + 20x + 20$.
    (b) Hence, factorize $5x^2 + 20x - 25$.

Checkpoint 3

Factorize $x^2 + 3x + 2$

x^2
x
x

2	
$+2$	
$+1$	



You have learnt more!

Go on!

Factorize the following expressions.

13. $x^2 + 4x + 3$

14. $y^2 + 8y + 7$

15. $a^2 + 6a + 5$

16. $b^2 + 12b + 11$

17. $c^2 + 5c + 6$

18. $d^2 + 7d + 10$

19. $p^2 + 8pq + 15q^2$

20. $m^2 + 12mn + 35n^2$

21. $a^2 + 13ab + 30b^2$

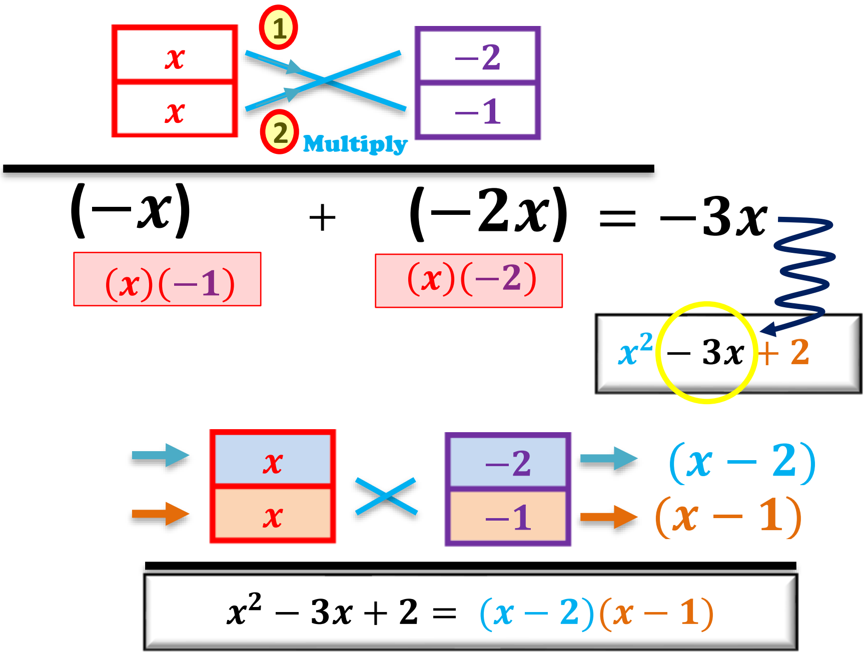
22. $x^2 + 17xy + 30y^2$

Checkpoint 4

Factorize $x^2 - 3x + 2$

x^2
x
x

2	
$+2$	-2
$+1$	-1



Keep Fighting!

Factorize the following expressions.

23. $x^2 - 14x + 13$

24. $y^2 - 13y + 12$

25. $w^2 - 8w + 12$

26. $z^2 - 7z + 12$

27. $p^2 - 11p + 28$

28. $q^2 - 16q + 28$

29. $r^2 - 9rs + 20s^2$

30. $a^2 - 18ab + 56b^2$

31. $s^2 - 19st + 70t^2$

32. $\alpha^2 - 16\alpha\beta + 48\beta^2$

Factorize the following expressions.

33. $a^2 + 5a + 6$

34. $a^2 + 5 + 6a$

35. $b^2 - 12b + 27$

36. $3b^2 - 36b + 81$

37. $x^2 - 15x + 36$

38. $2x^2 + 30x + 72$

39. $m^2 - 10mk + 21k^2$

40. $m^2n - 10mnk + 21k^2n$

41. $s^2 + 15st + 26t^2$

42. $26s^2 - 15st + t^2$

43. $5c^2 - 40cd + 75d^2$

44. $p^3 + 18p^2t + 45pt^2$

45. $u^2t^2 + 13uvt + 22v^2$

46. $x^4 - 16x^2y^2 + 63y^4$

Unit Education

Give it a try

Expand the following expressions.

$(a+1)(a+2)$ $\quad$ $(a-1)(a-2)$ $\quad$ $(a+1)(a-2)$

最尾係正，正正負負

最尾係負，正負正負

Checkpoint 5

Factorize x^2+2x-3

Trial 1

x^2
x
x

-3	
$+3$	-3
-1	$+1$

x	①	$+3$
x	② Multiply	-1

$(-x) + (3x) = 2x$

$(x)(-1)$ $\quad$ $(x)(+3)$

$x^2 + 2x + 3$

Trial 2

x^2
x
x

-3	
$+3$	-3
-1	$+1$

x	①	-3
x	② Multiply	$+1$

$(x) + (-3x) = -2x$

$(x)(+1)$ $\quad$ $(x)(-3)$

$x^2 + 2x + 3$

Final Ans:

$$x^2+2x-3=(x+3)(x-1)$$

Factorize the following expressions.

47. $x^2 + 2x - 3$

48. $x^2 - 2x - 3$

49. $y^2 + 3y - 10$

50. $y^2 - 3y - 10$

51. $h^2 - 14h - 51$

52. $n^2 + 16n - 57$

53. $r^2 - 8r - 33$

54. $k^2 + 3k - 54$

55. $x^2 - 11xy - 42y^2$

56. $a^2 - 9ab - 52b^2$

57. $c^2 + 19cd - 120d^2$

58. $p^2 + 14pq - 72q^2$

Final Boss!

Factorize the following expressions.

59. $c^2 - 28d^2 + 3cd$

60. $p^2 + 40q^2 - 14pq$

61. $-6st + s^2 - 91t^2$

62. $-r^2 + 17rs - 70s^2$

63. $-u^2 - 13uv + 68v^2$

64. $-68n^2 - 21mn - m^2$

65. $x^2 - 36y^2 - 9xy$

66. $3a^2 - 15ab - 42b^2$

67. $2p^2 - 19pq + 45q^2$

68. $3h^2 - 16hk - 35k^2$

69. $-119\theta\phi - 42\theta^2 + 98\phi^2$

70. $24\alpha^4\gamma^2 + 90\beta^4\gamma^2 - 94\alpha^2\beta^2\gamma^2$